

A service system for BT's major customers

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Customer service excellence is regarded as one of the key differentiators of BT's product and service portfolio. This paper discusses, from a technical perspective, the major customer service system (MCSS), and how it is helping to achieve excellent customer service for BT's global and major customers.

1. Introduction

This paper focuses on the technical challenges associated with the development and deployment of the major customer service system (MCSS) — a system designed to support BT's top 4500 global and major customers, who between them account for nearly one third of BT's total revenue.

The paper discusses some of the unique features of the MCSS architecture and illustrates how a diverse range of technologies from 4GL tools, three-tier client/server and Intranet-based network computing have been combined to provide a key competitive edge to BT's customer service capability.

The following three key requirements have dictated the architecture of MCSS.

- Electronic customer access (ServiceView)

MCSS users include both traditional service agents within the customer service centres responding to telephone enquiries, but also includes direct electronic customer access (with the attendant security issues). This allows customers to raise and track orders and problem reports without the need to directly involve BT agents.

- Product diversity

The range of products and services supported by MCSS ranges from PSTN and ISDN, through private circuits, ATM and 0800, and also encompasses global products such as Concert frame and packet services. Central to the external market positioning of ServiceView is the ability to support all of these services through an integrated management system, rather than building stand-alone systems to support specific products.

- Performance and scalability

MCSS is deployed in a large number of separate and distinct environments, from small systems supporting BT's joint ventures, up to the main UK customer deployment with the requirement to support 1700 users.

2. The business context for MCSS

MCSS is part of the major customer service excellence programme, the scope of which encompasses building uplifts, business process re-engineering and much more besides. However, from a systems perspective the business benefits of MCSS are delivered through three key capabilities.

Firstly, BT and its major customers can share information electronically as the basis of forging an outgoing customer relationship. This is realised through an integrated system supporting a wide range of information and transaction types (from order entry to bill enquiries), and encompassing the vast majority of BT's products and services portfolio. This electronic interaction and the associated software applications significantly improve the accuracy and timeliness of information flows between the customer and BT.

Secondly, customers in this market segment typically have over 5000 distinct BT products and services. By building a customer-centric information repository, BT has moved away significantly from the traditional support systems strategy where customer data has been distributed through a range of product-based systems.

This repository can now serve as the customer's main inventory system and is fundamental to supporting the major MCSS applications, which customers can access

through ServiceView. It also empowers the BT account support teams with the key information they need to start to become more proactive in anticipating and meeting their customer's needs.

Thirdly, there are operational cost savings, since a side effect of providing electronic customer access is that information can be automatically interchanged with the customer. However, if MCSS was just a stand-alone system this benefit would be marginal as information would still need to be entered into other BT systems; therefore a series of automated interfaces have been built that allow for this interchange of data. These interfaces will also reduce the need for training on a wide variety of complex legacy systems, with users instead only needing to learn the single graphic user interface for the MCSS.

3. Systems architecture

Figure 1 illustrates the logical architecture of MCSS, which is based on a 'traditional' three-tier model, i. e. Windows or Web clients, communicating with an open systems 'mid-tier' server, which in turn interacts with a variety of BT and global mainframe servers.

Customers interact with the global and major service centres via a variety of routes. Early customers used the Windows-based MCSS manager client; more recently customers are starting to use a World Wide Web (WWW) Intranet client (Bali). This latter direction has been inspired by the need to reduce the operational cost of delivering

electronic access to an ever-increasing customer base. Of course many customers still contact the service centres via telephone, but in addition to voice communication ServiceView customers are provided with a video link, to enable face-to-face dialogue with the service centre agents. Within BT the service agents currently use the MCSS manager client.

The MCSS server is based on an Oracle database, which in high-end deployments is configured as a parallel database running on clusters of multi-processor SPARC Enterprise servers. Currently MCSS has its own in-built operational customer (OCDB) and product (PDB) databases, but the intention is to migrate towards the emerging corporate systems as they become available.

MCSS supports integration with a large number of other systems, including the customer service system [1] (CSS), the customer-oriented system for the management of special services [2] (COSMOSS), the computer assisted maintenance for special services (CAMSS), OMNI and CTMS via its external systems interface architecture. A combination of server-based robotics and, increasingly, customer-handling intermediate server (CHIS) services [3] are used to facilitate this communication. The same architecture also supports the ability for separate MCSS systems to interwork; this multiserver working is used to provide scalability for large installations, plus the ability for different MCSS-based organisations to interwork, enabling capabilities such as 'follow the sun' working.

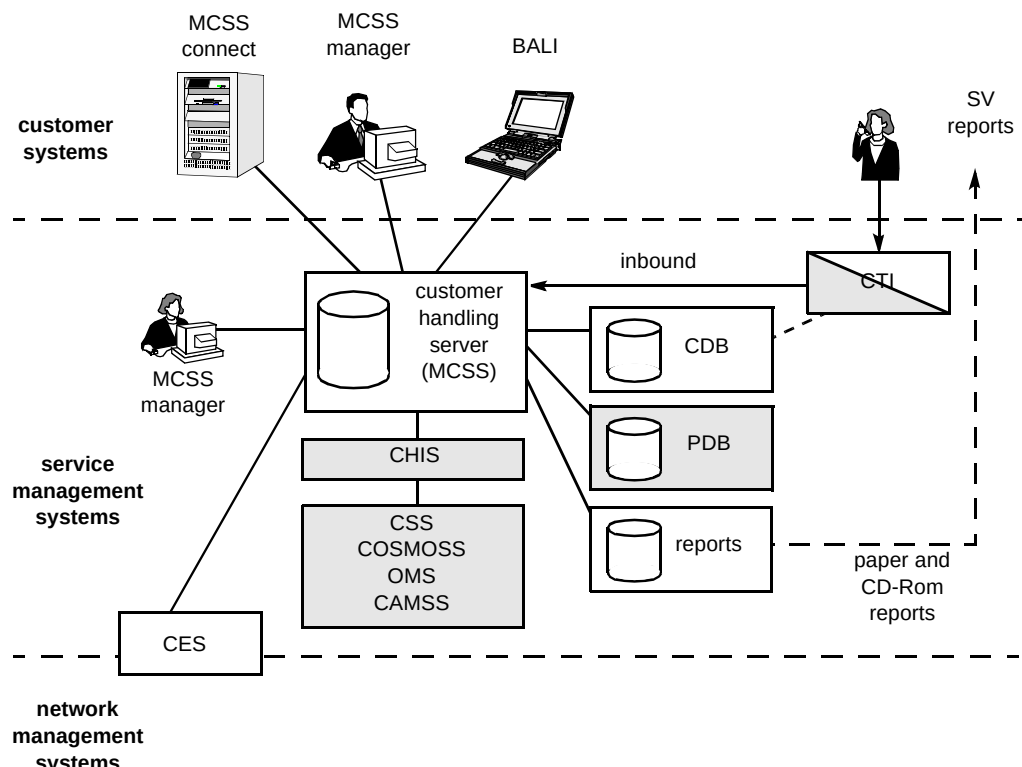


Fig 1 The MCSS logical architecture.

MCSS also handles feeds of network events from a variety of BT's network services, thus enabling proactive reporting of service-affecting faults to customers and service agents. A standard component (communications and event server (CES)), based on HP's OpenView platform, is used to provide this capability.

Regardless of the client 'type' or the 'back-end' systems supported, a standard application programming interface (API) provides access to all of the core server-based functionality of the MCSS system. This API is now also being used by third parties (e.g. BT's joint ventures and remote development teams) to extend and enhance the core application and interface set.

4. Functional architecture

MCSS applications have been developed using a highly modular and extensible approach, with a variety of distinct applications supported by a common infrastructure and data model (see Fig 2). The descriptions below provide a brief overview of some of the key components together with an insight into the approach taken for their construction.

4.1 Infrastructure and data

There are five key components of the MCSS infrastructure and data modules.

- Security

Because of the need to provide external customer access, and also to support the fact that operationally MCSS users are typically organised into distinct customer teams, security and data partitioning considerations have been paramount in the design.

MCSS supports 'strong' end-user authentication mechanism, full data encryption across the client/server links and low-level data partitioning based on an MCSS-developed (BT patented) algorithm. A primary characteristic of the design has been to provide the

security features transparently to the application developers, who therefore do not have to worry about the mechanics of the security regime. MCSS applications have been certified to European community standard E4.

- Workflow

MCSS was always designed to support a variety of operational deployments and a wide range of products. An early design decision therefore was to build all the applications around a finite state model, where the available 'states' and the legal 'state' transitions are defined by 'data-driven workstrings'. This approach has enabled the tailoring of MCSS around existing business processes, e.g. it is possible to define in data the order-taking process from initial data capture, through order validation, authorisation, and interaction with the appropriate order fulfilment system.

- External interfaces

External systems are treated just like any other client of the MCSS server; however, a series of API calls are available to facilitate integration with distributed computing environment (DCE) services such as CHIS.

The system also supports several 'standard' external interfaces such as bi-directional e-mail messages, facsimile and radio pager gateways. In addition, a server-based robotics infrastructure is available to support screen-scraping interfaces such as 3270 and VT100.

- Computer/telephony integration (CTI)

The MCSS programme currently uses CTI to enable its customer reception application. This facility will cause an automatic 'screen pop' of relevant customer details based on the calling and called telephone numbers. The application is based on the Genesys T-Server CTI

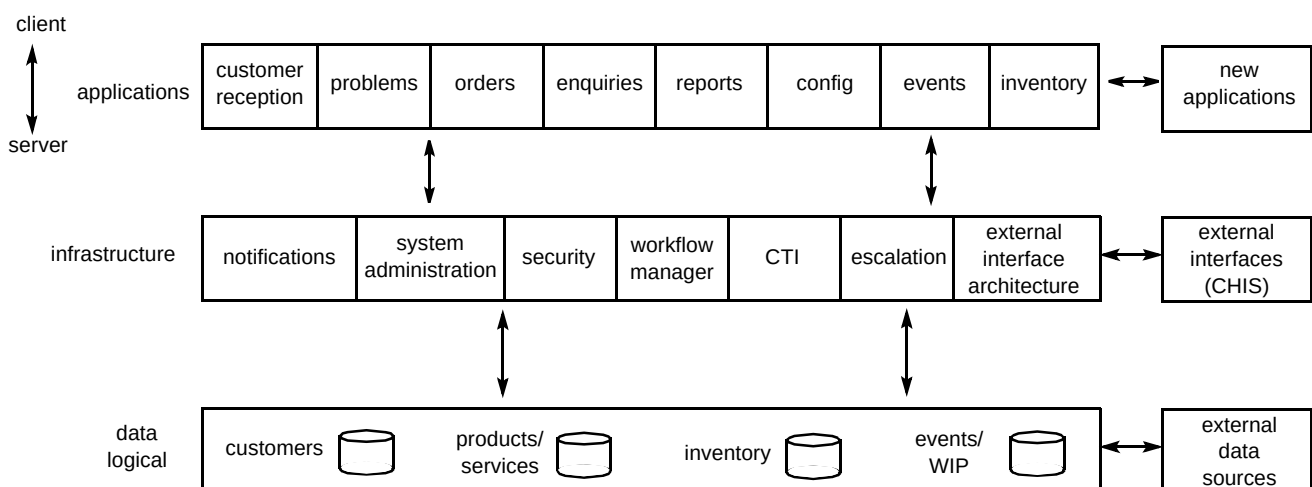


Fig 2 The MCSS functional architecture.

product, and is currently on trial as part of an extension of the scope of MCSS into the extended business customer base.

- Data

One of the fundamental design decisions in MCSS was to create a customer database at the mid-tier that replicated much of the data on the back-end systems. The key rationale for this approach was to create a consolidated customer-centric view of information that could be provided to both customers and to the customer teams. This information is a key part of the sales proposition of ServiceView, and is the most frequently cited reason for attracting customers to the ServiceView portfolio. It also begins to empower the customer teams with the information they need to move from a reactive to a proactive approach with their customers.

The customer data includes customer names, hierarchies, contacts and addresses, sites and locations, full product and service inventory, and work in progress (WIP) details, e.g. outstanding orders and faults.

4.2 Applications

The MCSS applications can be divided into both client and server functions, the server functions typically being realised as reusable business objects implemented in Oracle's PL-SQL programming language.

Although the client applications have traditionally been Oracle forms-based applications running on Windows PCs, many of these applications have now been rewritten as WWW applications in a combination of HTML, CGI and Java. There is an ongoing development programme to accelerate and extend the availability of the MCSS applications on the WWW.

- Customer reception

Customer reception is one of several entry point applications; it supports initial customer profiling and data capture, and then enables rapid navigation to the most appropriate application, based on the initial customer dialogue.

- Inventory

The inventory application provides the ability to browse and configure a customer's portfolio of products and services. Numerous tools are provided to interactively report on the inventory, using techniques such as query by example (QBE) and user-definable filters. Typically users might want to ask questions of the inventory database such as: 'How much bandwidth is available between sites A and B?', where that bandwidth may be realised by a variety of different services (e.g. flexible bandwidth service, kilostreams and megastreams).

All service inventory object classes contain a standard 'header' which identifies the object to the MCSS system. The details of a particular service are normally contained in an inventory 'body', supplied as a distinct executable (analogous to a Windows dll), or alternatively (and more typically) by one of several 'meta' service bodies. In this latter case, all of the attributes, screen layout and validation rules are defined in a server-side product database, with no need to modify the client software.

- Order processing

The order-processing application is responsible for capturing order details, updating the MCSS inventory and then, via the workflow engine, forwarding the order requests to the appropriate external system. Figure 3 shows the order entry screen.

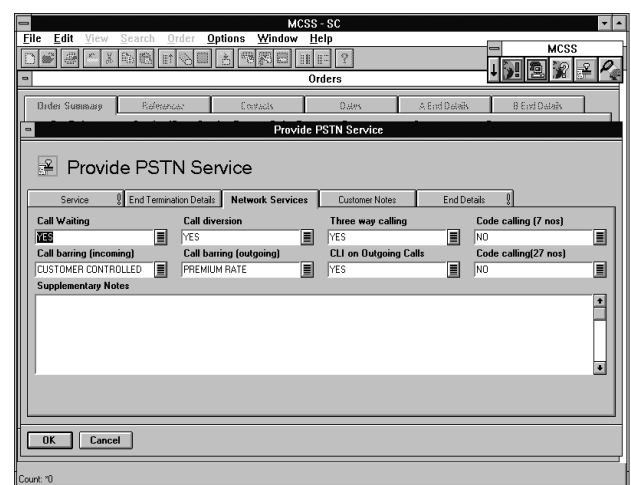


Fig 3 The order entry screen.

The same concepts of order header and order body described above are reused in this application. In addition, an order 'installation' routine can be supplied which determines how fulfilled orders are added to the inventory database.

- Fault processing

The fault-processing functions also operate in a similar manner. It should be noted that faults can also be automatically raised via CES or an external system and can then be automatically forwarded to the customer's ServiceView terminal.

The workflow engine is sufficiently powerful to control all of the key state transitions within the fault and order-processing applications, and is also responsible for managing inter-workgroup communications and automatically informing customer systems and other BT systems of key events and progress under the control of the data-driven workstrings that represent the business process model. Workflow can also be integrated with a prompting engine that allows

time-out 'profiles' to be associated with workstrings, and the appropriate escalations to be triggered, e.g. if an application remains in any given state too long.

In both the fault and order-processing applications it is possible to track progress, both within MCSS and also on the external systems as key events are delivered across the automated interfaces.

- Configuration

Configuration tools are a classic example of the occasional need to supply product-specific applications to supplement the more general-purpose ones available from the 'core' MCSS product. One of the more ambitious of these applications has been developed for BT's telemarketing services (0800, 0345, etc).

Figure 4 illustrates a call plan editor application that allows BT's telemarketing customers to manipulate their call routing plans with an almost immediate response time, enabling campaign routing to be tuned graphically against a wide range of possible decision points dependent on numerous factors from date and time of day, to which region of the country is generating the call.

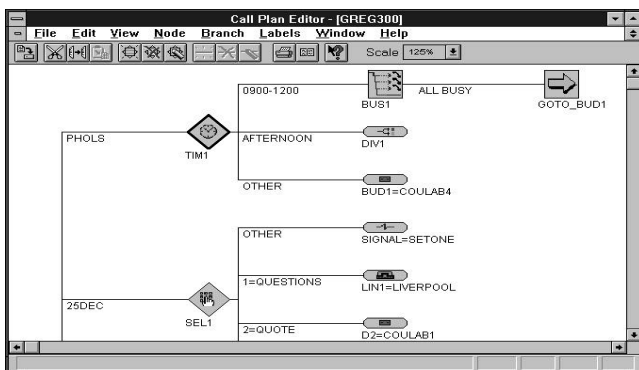


Fig 4 The call plan editor screen.

- Shared workspace

Shared workspace is a feature supported only by the WWW clients; it basically provides a jointly owned and maintained information pool shared between BT and the customer. Account team contacts, key diary events, customer service plan details are held on-line; 'content' management tools are supplied to enable end users to manage the information directly and therefore to keep it up to date. Figure 5 shows the shared workspace signpost.

4.3 The key construction challenges

At the outset of the development stages for MCSS in 1994, it was decided to use a 4GL toolkit rather than an in-house developed platform, utilising the Oracle Designer and Developer 2000 products with the Oracle database. This has enabled a significant amount of the standard functionality to

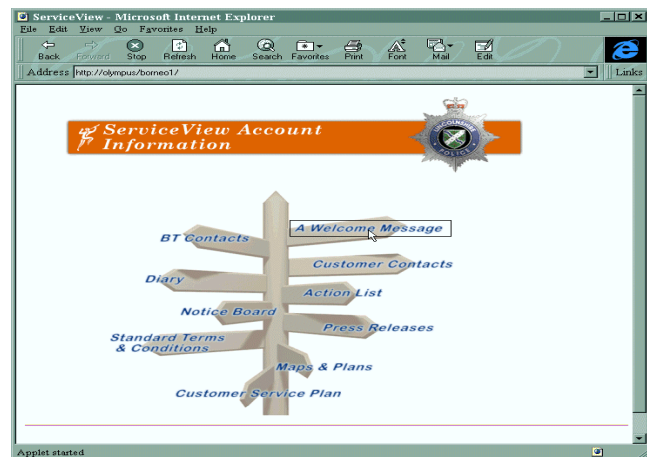


Fig 5 The account information signpost screen.

be exploited, such as encryption, multiple-platform support, scalability and password management. Further evolution will make use of the Oracle Web Server, Sedona and Oracle 8 products, as well as the common object request broker architecture (CORBA).

Of paramount importance in the deployment of a system is performance, and the key area is the finite network pipe between the client and the server where indiscriminate and unnecessary data transfer costs user response time. This desire to minimise network bandwidth between client and server, especially for WWW clients, led to the decision to encapsulate functionality behind a server-side API, which produced all the results prior to returning them to the client.

The decision to maintain a customer-centric database has been described earlier in this paper. This feature is central to the MCSS functionality; however, replicating and managing up to 30 Gbytes of data sourced from at least six external system sources potentially leads to a variety of data quality and integrity issues.

The MCSS solution to managing these issues has been the creation of a customer data management centre (CDMC), where the approach has been the development of a data-cleanse tool to enable the correction of data at the source systems, using techniques such as PAF (Post Office address file) matching and de-duplication, prior to loading on to the MCSS. This also has the advantage of improving the data for the native external system users. While the MCSS order and configuration applications should then maintain synchronisation with the external systems via automated interfaces, the CDMC will also supply a 'routing' function to periodically check data integrity with MCSS.

Deploying and managing large client/server systems is always going to be a massive undertaking. Now there is the necessary tools support for software distribution and systems management in the client/server environment, e.g. the NetView/6000-based enterprise systems management

facility (ESMF), into which MCSS has been integrated via the generation of SNMP traps.

Finally, another fairly unique requirement of MCSS is multi-language deployments. The strategy to date for multiple-language support is restricted to help file changes or alternatively a specific language 'port' of the software. Here, again, the WWW variant of MCSS may provide a true solution to multi-language support without the need for client-side software changes.

5. The future

Intranet-based applications are the key to the rapid and widespread deployment of MCSS to the high-end business customer market-place, and the MCSS programme is already delivering this type of application to external customers and starting to reap the benefits in operational cost savings and speed to market. Internet-based applications are on their way and will enhance BT's ability to extend the functions of MCSS to a much broader customer base than would otherwise have been cost effective.

Acknowledgements

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He is currently the chief designer of BT's portfolio of service, marketing and sales systems.